Human Capital

Raffaele Saggio

Econ 560

September 8, 2026

Human Capital Investments: Theory and Evidence

- Mincerian earnings regression:

$$\log y_i = \alpha + \beta S_i + \gamma_1 Exp_i + \gamma_2 Exp_i^2 + e_i$$

- Questions:
  - What is the theoretical rationale for this model?
  - Does β have a causal interpretation? If not, is it too big or too small?
  - What do we know about the causal effect of schooling on earnings?

Investment Over the Life Cycle: The Ben-Porath Model

- Human capital paradigm: Worker skills as a form of capital
- Workers invest in these skills to improve earnings opportunities
    - Schooling
    - On-the-job training
- Tradeoff: cannot work while investing in human capital
- Elegantly described by Ben-Porath (1967)

A Causal Model of Schooling Investments

- To investigate the returns to education, consider a static model of schooling investment, as in Card (1999) Handbook Chapter (almost 7K citations!).
- Consider an individual deciding how long to stay in school, S.
- Schooling maps to earnings according to the differentiable function $y(S)$ where $y'(S) \equiv \dfrac{\partial y(S)}{\partial S}$ denote the marginal return of schooling.
- An individual's goal is to maximize the present discounted value of earnings ($y(S)$) given a discount rate of r.

$$\max_{S} PDV(S)$$

  where $PDV(S) \equiv \int_{S}^{\infty} e^{-rt} y(S) dt$.

- Idea: individuals discount future earnings at rate r and decide how many years to spent in school before earning some amounts of money.
- What are we implicitly assuming here?

A Causal Model of Schooling Investments

- Note that the integral can be written as:

$$\int_S^{\infty} e^{-rt} y(S) dt = -y(S) \frac{e^{-rt}}{r} |_S^{\infty} = y(S) \frac{exp(-rS)}{r}$$

- Therefore the FOC of the maximization problem of the previous slide is:

$$y'(S^*) \frac{exp(-rS^*)}{r} - y(S^*) exp(-rS^*) = 0$$

which can be written as

$$\frac{dlogy(S^*)}{dS} = r$$

- Intuition?

A Causal Model of Schooling Investments

$$\frac{dlogy(S^*)}{dS} = r$$

- Useful framework for interpreting empirical evidence and thinking about heterogeneity
- Variation in schooling across individuals is driven either by heterogeneity in the earnings function $y(S)$ or the interest rate r
- "Ability bias:" Individuals that choose different schooling levels may have different potential earnings functions, leading observed earnings/schooling relationship to differ from causal impact of schooling

Ability Bias

- Simple specification of earnings function:

$$y_i(S) = \alpha_i + \beta S_i$$

- Optimal schooling choices: $S_i^* = \frac{1}{r} - \frac{\alpha_i}{\beta}$
- Are "smart" dudes getting more or less schooling in this model?
- What coefficient would a regression of earnings on schooling (and a constant) return?
- Is OLS biased upwards or downwards?

Ability Bias

- Alternative specification of earnings function:

$$logy_i(S) = \alpha_i + \beta S_i - \frac{\gamma}{2} S_i^2$$

- What is different now?
- Assume also that discount rates vary across individuals r_i and that r_i has a symmetric distribution.
- Optimal level of schooling

$$S_i^* = \frac{\beta - r_i}{\gamma}$$

- OLS regression of log earnings on schooling:

$$b = \frac{Cov(\log y_i(S_i^*), S_i^*)}{Var(S_i^*)}$$

Ability Bias

- Since r_i has a symmetric distribution with variance denoted by σ_r^2 and covariance with α_i given by $\sigma_{\alpha,r}$, we have that

$$b = \beta - \gamma E[S_i^*] - \gamma \frac{\sigma_{\alpha,r}}{\sigma_r^2}$$

- First term is population average return to schooling
- Second term is the bias – depends on joint distribution of log earnings intercept α_i and cost r_i
- What do you think is the sign of $\sigma_{\alpha,r}$? Is OLS biased downwards or upwards?

Returns to Schooling Empirics

- Equation of interest:

$$\log y_i = \alpha + \beta S_i + \gamma_1 Exp_i + \gamma_2 Exp_i^2 + e_i$$

- Why adding controls for experience?

Returns to Schooling Empirics

- Equation of interest:

$$\log y_i = \alpha + \beta S_i + \gamma_1 Exp_i + \gamma_2 Exp_i^2 + e_i$$

- Why adding controls for experience?
- Facts:
  - OLS return β is 7 to 8 percent in most data sets
  - Empirical relationship between log earnings and schooling is surprisingly linear
  - Additively separable quadratic experience profile fits the data well

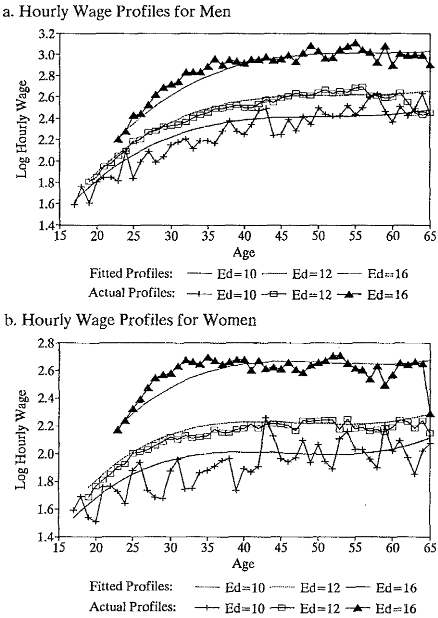


Fig. 1. Age profiles of hourly wags for men (a) and women (b).

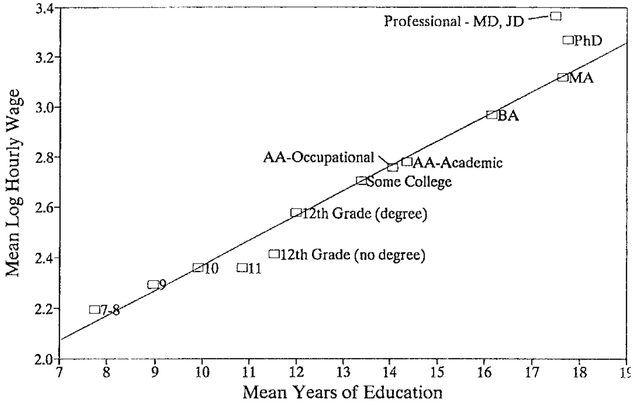


Fig. 2. Relationship between mean log hourly wages and completed education, men aged 40–45 in 1994–1996 Current Population Survey. Mean education by degree category estimated from February 1990 CPS.

Angrist and Krueger (QJE 1991)

- Angrist and Krueger (1991) seek to estimate the causal return to education
- Instrumental variables strategy motivated by interaction between compulsory schooling and age-at-entry laws
  - Students can typically drop out on the day they turn 16
  - Birth date cutoff for starting age: Usually start in the fall of the calendar year in which they turn six
- Generates differences in mean attainment by date of birth

Birth date	School start date	Dropout date	Schooling at dropout date
January 2, 1930	September 1, 1936	January 2, 1946	9.5 years
December 31, 1930	September 1, 1936	December 31, 1946	10.5 years

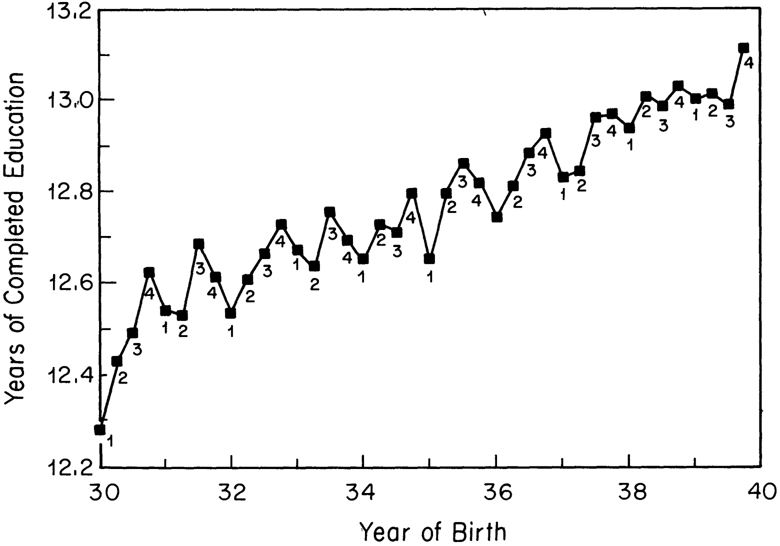


FIGURE I
Years of Education and Season of Birth
1980 Census
Note. Quarter of birth is listed below each observation.

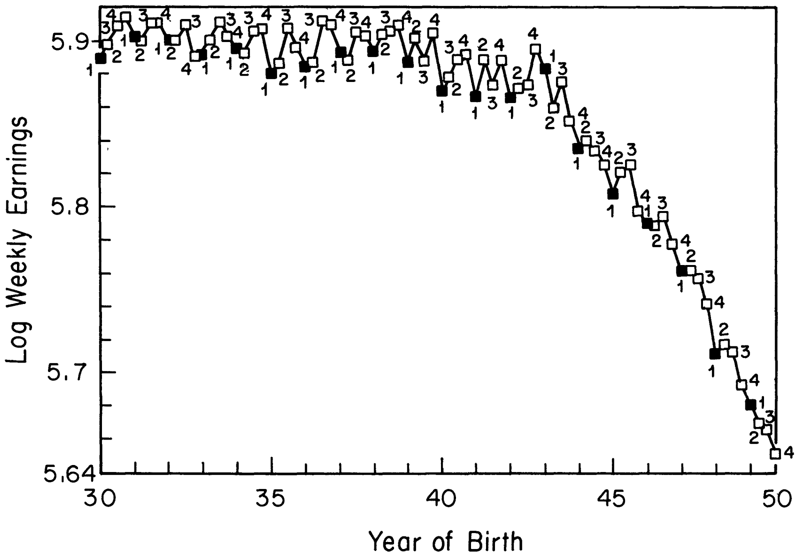


FIGURE V
Mean Log Weekly Wage, by Quarter of Birth
All Men Born 1930–1949; 1980 Census

Panel B: Wald Estimates for 1980 Census—Men Born 1930–1939

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) − (2)
ln (wkly. wage)	5.8916	5.9027	−0.01110
			(0.00274)
Education	12.6881	12.7969	−0.1088
			(0.0132)
Wald est. of return to education			0.1020
			(0.0239)
OLS return to education			0.0709
			(0.0003)

IV vs. OLS

- AK find IV estimates at least as large as OLS – seems surprising
- Card (1999): same pattern for other instruments
- Potential explanations?

IV vs. OLS

- AK find IV estimates at least as large as OLS – seems surprising
- Card (1999): same pattern for other instruments (can you think of some instruments for schooling?)
- Potential explanations:
  - Pure opportunity cost
  - Measurement error
  - Publication bias / File Drawer Problem
  - IV estimates a local average treatment effect for a particular group of individuals known as compliers (we will come back to this in Econ 628).

Challenges to QOB

- Challenges to validity and interpretation of QOB estimates?

Challenges to QOB

- What do you think are the challenges to the interpretation of these IV estimates?
- We will now review some more recent estimates of the returns to education (most of them are based on a RD design).

Challenges to QOB

- What do you think are the challenges to the interpretation of these IV estimates?
- We will now review some more recent estimates of the returns to education (most of them are based on a RD design).

Compulsory Schooling Laws in the UK

- More recent estimates of the return to schooling: studies of compulsory schooling reforms in the UK
- In 1947, legal school-leaving age in Britain went from 14 to 15
- Children born before April 1933 could leave school on their 14th birthdays; children born later could not leave until 15
- This change affected nearly half the population
- Oreopolous (2006) uses this change in an RD design, finds large returns
- Later challenged/updated by Devereux and Hart (2010)
- Grenet (2013) studies a similar change from 15 to 16 in 1972

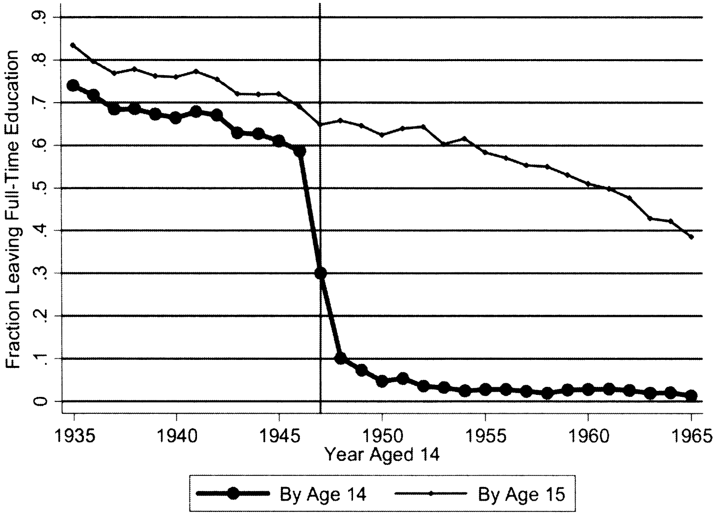


FIGURE 1. FRACTION LEFT FULL-TIME EDUCATION BY YEAR AGED 14 AND 15
(*Great Britain*)

Note: The lower line shows the proportion of British-born adults aged 32 to 64 from the 1983 to 1998 General Household Surveys who report leaving full-time education at or before age 14 from 1935 to 1965. The upper line shows the same, but for age 15. The minimum school-leaving age in Great Britain changed in 1947 from 14 to 15.

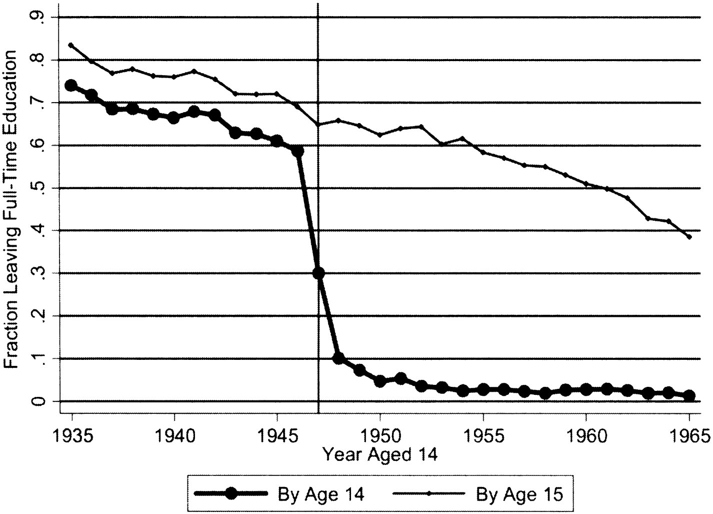


FIGURE 1. FRACTION LEFT FULL-TIME EDUCATION BY YEAR AGED 14 AND 15
(*Great Britain*)

Note: The lower line shows the proportion of British-born adults aged 32 to 64 from the 1983 to 1998 General Household Surveys who report leaving full-time education at or before age 14 from 1935 to 1965. The upper line shows the same, but for age 15. The minimum school-leaving age in Great Britain changed in 1947 from 14 to 15.

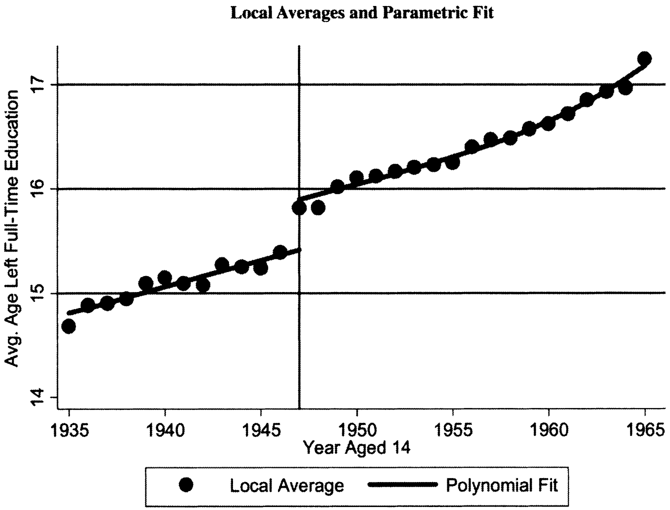


FIGURE 4. AVERAGE AGE LEFT FULL-TIME EDUCATION BY YEAR AGED 14
(*Great Britain*)

Note: Local averages are plotted for British-born adults aged 32 to 64 from the 1983 to 1998 General Household Surveys. The curved line shows the predicted fit from regressing average age left full-time education on a birth cohort quartic polynomial and an indicator for the school-leaving age faced at age 14. The school-leaving age increased from 14 to 15 in 1947, indicated by the vertical line.

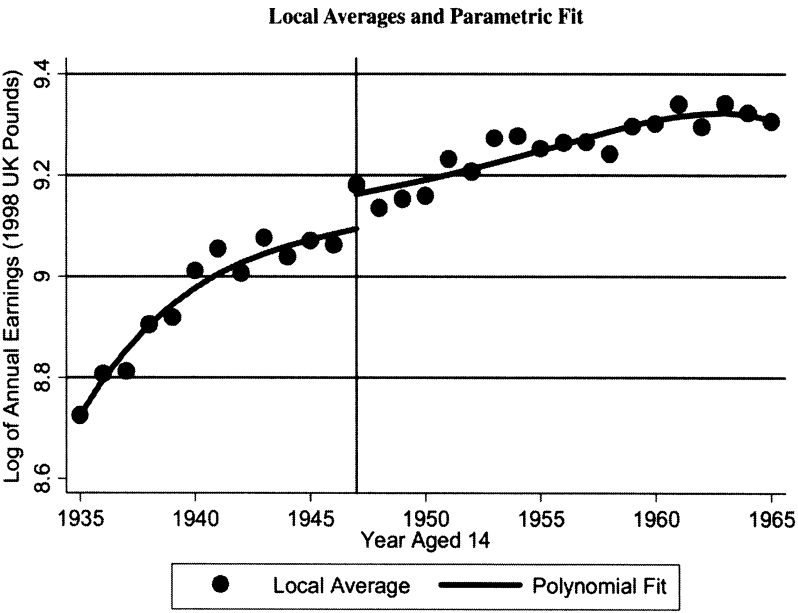


FIGURE 6. AVERAGE ANNUAL LOG EARNINGS BY YEAR AGED 14
(*Great Britain*)

TABLE 2—OLS AND IV RETURNS TO (COMPULSORY) SCHOOLING ESTIMATES FOR LOG ANNUAL EARNINGS
(Great Britain and Northern Ireland, ages 25–64, 1935–1965)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Returns to schooling: OLS			Returns to compulsory schooling: IV			Initial sample size
Great Britain	0.078	0.079	0.079	0.147	0.145	0.149	57264
	[0.002]***	[0.002]***	[0.002]***	[0.061]**	[0.063]**	[0.064]**	
Northern Ireland	0.111	0.113	0.113	0.135	0.187	0.21	8921
	[0.004]***	[0.004]***	[0.004]***	[0.071]*	[0.070]**	[0.135]	
G. Britain and N. Ireland with	0.082	0.082	0.083	0.174	0.149	0.148	66185
N. Ireland fixed effect	[0.001]***	[0.001]***	[0.001]***	[0.042]***	[0.044]***	[0.046]***	
Birth cohort polynomial controls	Quartic	Quartic	Quartic	Quartic	Quartic	Quartic	
Age polynomial controls	None	Quartic	None	None	Quartic	None	
Age dummies	No	No	Yes	No	No	Yes	

Notes: The dependent variable is log annual earnings. Each regressions includes controls for a birth cohort quartic polynomial and age left full-time education (instrumented by an indicator whether a cohort faced a school leaving age of 15 at age 14 in columns 4 through 6). Columns 2, 3, 5, and 6 also include age controls: a quartic polynomial and fixed effects where indicated. Each regression includes the sample of 25- to 64-year-olds from the 1983 through 1998 General Household Surveys who were aged 14 between 1935 and 1965. Data are first aggregated into cell means and weighted by cell size. Regressions are clustered by birth cohort and region (Britain or N. Ireland).

Devereaux and Hart 2010

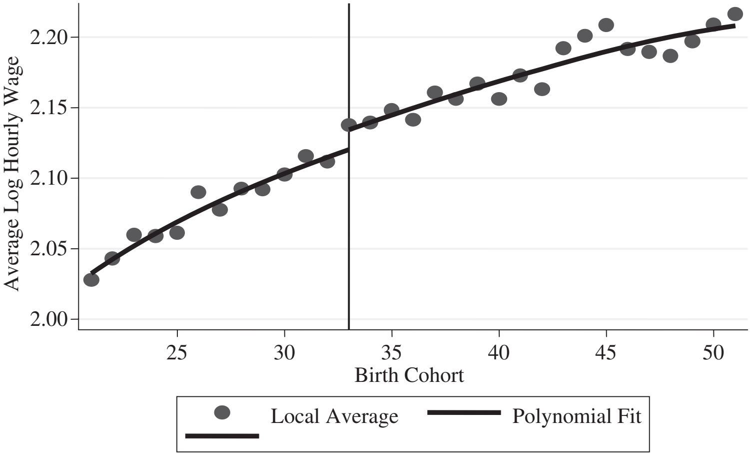


Fig. 8. *Average Hourly Wage by Year-of-Birth for Men* (NESPD)

Devereaux and Hart 2010

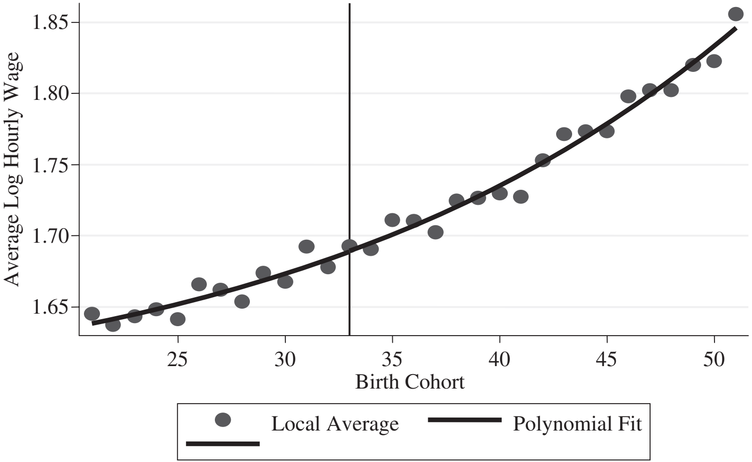


Fig. 7. *Average Hourly Wage by Year-of-Birth for Women* (NESPD)

Table 3. *Continued*

Estimation method	OLS	First-stage		Reduced-form		
Dependent variable	Log of hourly wage	School-leaving age		Log of hourly wage		Employed
	(1)	(2)	(3)	(4)	(5)	(6)
Panel B: England and Wales, school cohorts 1949−1967 (Quarterly Labour Force Survey 1993−2006)						
Women						
Coefficient	0.119***	0.309***	0.310***	0.019***	0.021***	0.007**
(s.e.)	(0.001)	(0.023)	(0.022)	(0.003)	(0.003)	(0.003)
Number of observations	75,304	75,304	75,304	75,304	75,304	129,813
Men						
Coefficient	0.095***	0.267***	0.267***	0.019**	0.019**	0.002
(s.e.)	(0.001)	(0.028)	(0.028)	(0.008)	(0.008)	(0.004)
Number of observations	68,535	68,535	68,535	68,535	68,535	117,627
Age controls	Quartic	None	Quartic	None	Quartic	Quartic

Notes: ***, **, and * denote significance at the 1, 5, and 10 percent levels, respectively. Each coefficient comes from a separate
hourly wage (Columns 1, 4, 5, 8, and 9), age of leaving full-time education (Columns 2 and 3), an indicator variable for being
for being self-employed (Column 7). OLS regressions in Column 1 include a quartic in age and a full set of survey year dumm
fourth-order polynomial in school cohort, allowing for an intercept shift at the cut-off point (i.e., 1953 in France and 1958 in En
is constructed from the first wave of the 1990–2002 French Enquête Emploi. It includes individuals who were born in France, wh
full-time education between the ages of 6 and 25, and who were aged 25–60 when surveyed. The sample from England and Wa
of the 1993–2006 Quarterly Labour Force Survey. It includes individuals who were born in England and Wales, who belong to
September 1948 and 31 August 1967), who left full-time education between the ages of 6 and 25, and who were aged 25–60
include all individuals. The subsamples used in Column 7 are restricted to workers. The subsamples used in Columns 1–5 and

Returns to College for Marginal Students: Zimmerman (JOLE 2014)

- Much of the literature on returns to schooling focuses on students on the margin of dropout in mid-20th century
- Perhaps more policy-relevant: Return for current marginal college students
- Zimmerman (2014) uses a combined SAT/GPA cutoff to estimate earnings effects at Florida International University (part of the Florida State University System–SUS)

Table 2
Sample Description

	All	FIU Applicant Sample	Marginal Applicant Sample	Labor Force Sample
White	.40	.18	.15	.15
Black	.27	.26	.33	.32
Hispanic	.28	.50	.47	.48
Male	.48	.37	.36	.35
Free or reduced-price lunch	.40	.43	.46	.46
High school GPA	2.63	2.92	2.72	2.72
SAT	NA	943	841	839
Attend SUS school next year	.16	.59	.51	.51
Attend community college next year	.31	.37	.50	.51
Survey: attend non-Florida college	.06	.04	.03	.03
Survey: attend Florida private college	.04	.08	.06	.06
In labor force sample	.68	.78	.80	1.00
Fraction quarters with earnings observations	.53	.64	.67	.83
N	351,198	24,690	8,147	6,542

NOTE.—FIU = Florida International University; SUS = Florida State University System. Sample means for selected student populations. FIU applicant sample refers to all FIU applicants. Marginal applicant sample refers to marginal FIU applicants. Labor force sample refers to marginal FIU applicants for whom outcome period earnings data are available. Fraction quarters with earnings observations is the fraction of quarters during the outcome period with uncensored (positive) earnings observations.

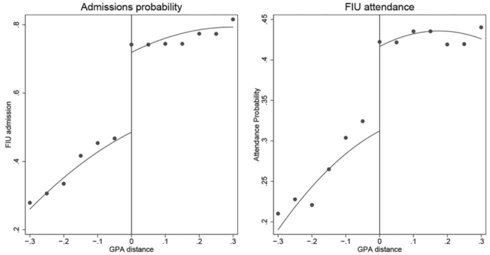


FIG. 4.—Admissions and FIU attendance. Lines are fitted values based on the main specification. Dots, shown every .05 grade points, are rolling averages of values within .05 grade points on either side that have the same value of the threshold-crossing dummy.

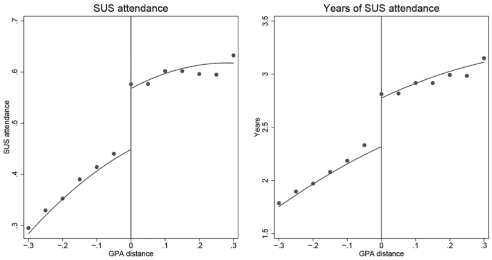


FIG. 5.—SUS attendance and persistence. Lines are fitted values based on the main specification. Dots, shown every .05 grade points, are rolling averages of values within .05 grade points on either side that have the same value of the threshold-crossing dummy.

Table 4: Effects on academic outcomes

Dep. Var.	Main	Controls	BW=0.5	BW=0.15	Loc. Lin.
A. Admissions and attendance					
Admitted to FIU	0.234***	0.233***	0.246***	0.282***	0.205***
	(0.021)	(0.018)	(0.022)	(0.023)	(0.016)
Attend FIU	0.104***	0.105***	0.112***	0.0980**	0.088**
	(0.025)	(0.026)	(0.029)	(0.040)	(0.027)
Attend SUS	0.119***	0.118***	0.126***	0.125**	0.104***
	(0.021)	(0.023)	(0.025)	(0.037)	(0.023)
Years SUS	0.457**	0.463**	0.492**	0.495**	0.420*
	(0.089)	(0.094)	(0.097)	(0.114)	(0.103)
SUS FTE terms	0.644*	0.643*	0.698*	0.650*	0.622
	(0.179)	(0.192)	(0.190)	(0.185)	(0.207)
B. SUS Graduation					
Within 4 years	-0.007	-0.008	-0.008	-0.009	-0.005
	(0.007)	(0.007)	(0.007)	(0.009)	(0.008)
Within 5 years	0.002	0.001	0.008	-0.002	0.007
	(0.018)	(0.019)	(0.018)	(0.021)	(0.021)
Within 6 years	0.057	0.057	0.056	0.044	0.069
	(0.022)	(0.022)	(0.026)	(0.022)	(0.024)
C. Other academic outcomes					
Years CC	-0.172*	-0.171*	-0.222**	-0.199**	-0.164*
	(0.053)	(0.051)	(0.067)	(0.055)	(0.061)
CC FTE terms	-0.338***	-0.327**	-0.394**	-0.412**	-0.300**
	(0.081)	(0.081)	(0.103)	(0.101)	(0.095)
AA within 6 years	-0.009	-0.005	0.005	-0.006	-0.001
	(0.021)	(0.020)	(0.021)	(0.021)	(0.025)
VC within 6 years	-0.007	-0.007	-0.006	-0.009	-0.006
	(0.006)	(0.006)	(0.005)	(0.003)	(0.006)
Survey: Out of state college	0.006	0.007	0.007	0.008	0.005
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Survey: In-state private	-0.012	-0.014	-0.010	-0.021	-0.009
	(0.009)	(0.009)	(0.009)	(0.014)	(0.011)
N	6542	6542	9659	3294	6542

Figure 8: Quarterly earnings by distance from GPA cutoff

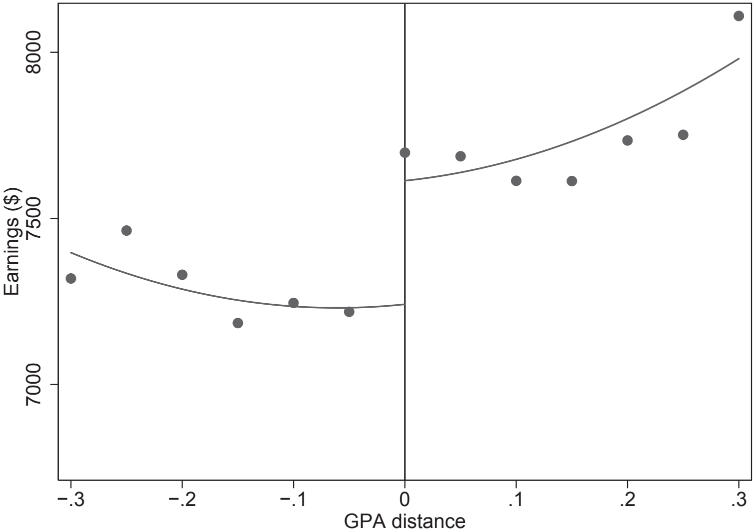


Table 5: Earnings effects 8 to 14 years after high school completion

	Main	Controls	BW=0.5	BW=0.15	Loc. Lin.
Reduced form estimates					
Above cutoff	372*	366**	409**	479**	410**
	(141)	(130)	(154)	(198)	(147)
Instrumental variables estimates					
FIU admission	1593*	1575**	1665**	1700**	2001*
	(604)	(584)	(645)	(621)	(696)
Years of SUS attendance	815**	792**	833**	966***	977**
	(276)	(262)	(271)	(305)	(306)
BA degree	6547*	6442*	7366*	10769	5958**
	(2496)	(2411)	(2998)	(5726)	(2024)
N	6542	6542	9659	3294	6542

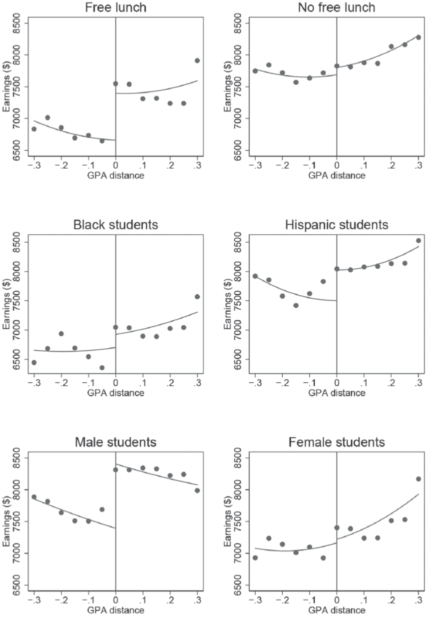


FIG. 9.—Heterogeneity in earnings effects. Lines are fitted values based on the main specification. Dots, shown every .05 grade points, are rolling averages of values within .05 grade points on either side that have the same value of the threshold-crossing dummy.

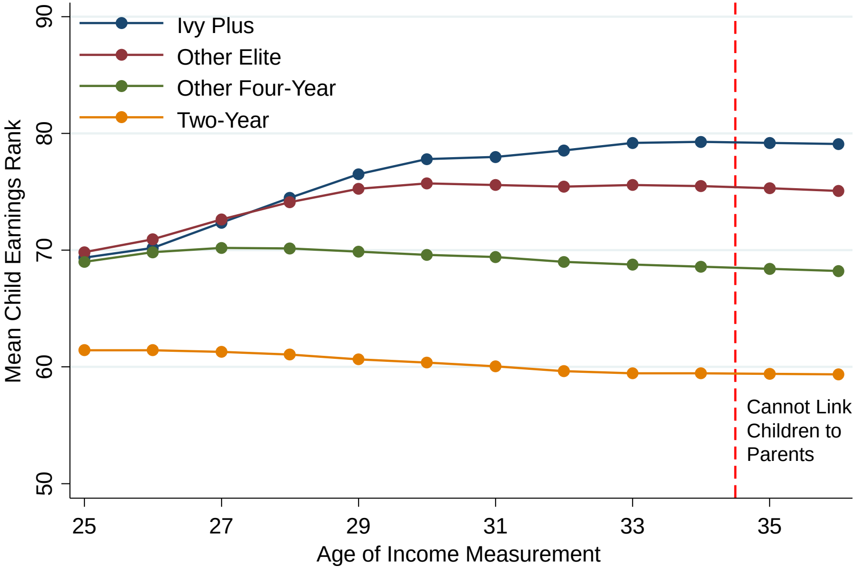
Table 6
Heterogeneous Effects in Educational Outcomes and Earnings

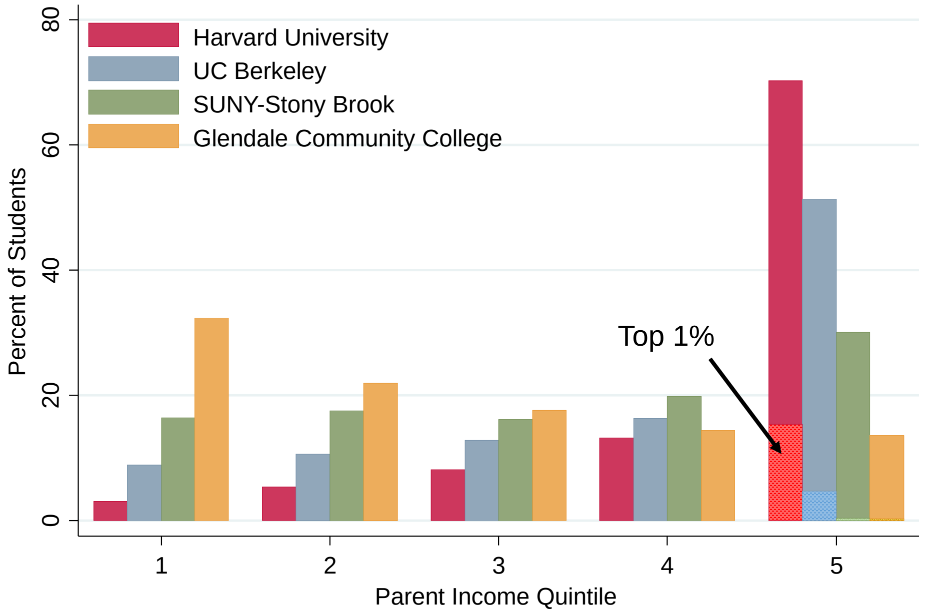
	Sample					
	Black	Hispanic	Male	Female	Free or Reduced-Price Lunch	No Free or Reduced-Price Lunch
A. Educational outcomes:						
FIU admitted	.276***	.233***	.242***	.230***	.274***	.212***
	(.031)	(.032)	(.040)	(.017)	(.037)	(.023)
Attend SUS	.140*	.118***	.102**	.129***	.140***	.111*
	(.040)	(.040)	(.027)	(.027)	(.037)	(.032)
Years SUS	.463	.394**	.477**	.436**	.477***	.474*
	(.178)	(.108)	(.149)	(.110)	(.104)	(.128)
BA in 6 years	.055	.063	.081	.043	.054	.063
	(.033)	.021	(.034)	(.018)	(.026)	(.024)
Years CC	−.245	−.098	−.261	−.129	.010	−.336
	(.128)	(.102)	(.143)	(.076)	(.116)	(.111)
AA in 6 years	−.047	.029	.011	−.020	−.006	−.012
	(.026)	(.047)	(.027)	(.023)	(.038)	(.024)
B. Earnings regressions:						
Reduced form	224	524*	1012**	56	737**	114
	(227)	(224)	(230)	(211)	(171)	(199)
IV: Admit	811	2255*	4191*	244	2695***	539
	(792)	(914)	(1324)	(916)	(521)	(940)
N	2,123	3,148	2,261	4,281	2,989	3,553

NOTE.—FIU = Florida International University; SUS = Florida State University System; BA = bachelor's degree; CC = community college; AA = associate's degree; IV = instrumental variable. Standard errors are clustered within grade pins. The *p*-values are calculated using a clustered wild bootstrap-*t* procedure described in Sec. III and app. B. All estimates are computed using the main specification above.
* Significant at the 10% level.
** Significant at the 5% level.
*** Significant at the 1% level.

Returns to College Selectivity

- For many students the relevant choice margin is which college to attend rather than years of schooling or college vs. no college
- Very large differences in earnings between students attending different US colleges, but also a huge amount of selection into college choice
- Hard to find credible experiments/quasi-experiments that induce variation in attendance at more vs. less selective colleges


Ivy Plus
Other Elite
Other Four-Year
Two-Year
Cannot Link
Children to
Parents
Mean Child Earnings Rank
Age of Income Measurement
90
80
70
60
50
25
27
29
31
33
35



Harvard University
UC Berkeley
SUNY-Stony Brook
Glendale Community College
Top 1%
Percent of Students
0
20
40
60
80
1
2
3
4
5
Parent Income Quintile

Returns to College Selectivity: Dale and Krueger

- Dale and Krueger (QJE 2002, JHR 2014) seek to provide some evidence on the returns to college selectivity
- Research design: compare students who applied to, and were admitted by, the same colleges, but chose to attend different ones
- Intuition: Application choices capture a lot of students' information about their own ability, while admission decisions capture a lot of colleges' information about student ability
- A student who was accepted by UBC and chose to attend SFU may better capture UBC students' "SFU counterfactual" than a random SFU student
- Note: this approach is fundamentally based on comparisons of individuals who make different choices

Dale and Krueger Approach 1

- Let $j \in \{1...J\}$ index colleges, and i index students
- C_{ij} indicates that student i chose to apply to school j, and A_{ij} indicates admission to j
- Collect in vectors $C_i = (C_{i1}...C_{iJ})$, $A_i = (A_{i1}..A_{iJ})$
- $D_i \in \{1...J\}$ denotes school attended by i
- Define Y_{ij} to be i's potential earnings if s/he attends college j. Observed earnings are $Y_i = \sum_j 1\{D_i = j\} Y_{ij}$
- DK's first approach is based on the assumption:

$$(Y_{i1}...Y_{iJ}) \perp\!\!\!\perp D_i | (C_i, A_i)$$

Dale and Krueger Approach 1

$$(Y_{i1} \ldots Y_{iJ}) \perp\!\!\!\perp D_i | (C_i, A_i)$$

- College choice is ignorable (independent of potential outcomes) conditional on application choices and admission outcomes
- Implies school choices are as good as random conditional on (C_i, A_i), so

$$E[Y_i | D_i = j, C_i = c, A_i = a] - E[Y_i | D_i = k, C_i = c, A_i = a]$$

$$= E[Y_{ij} - Y_{ik} | C_i = c, A_i = a]$$

- Within an application/admission portfolio, comparisons of mean observed outcomes give average treatment effects
- Note that such comparisons will be unavailable for most schools. DK aggregate comparisons of schools with higher/lower average SAT scores
- DK is a great example of a (credible) research based on a conditional independence assumption (CIA) $\rightarrow$ we will review this in more detail in Econ 628.

Dale and Krueger Approach 2

- Second approach: Control for average SAT score of the portfolio of schools to which a student applied
- DK describe this as a "self-revelation" model: causal interpretation requires that all selection is captured by students' decisions about how selectively to apply. No further information about potential outcomes in portfolio composition, admission decisions, or attendance choices
- Stronger assumption than approach 1, but allows for a more parsimonious model and more precision
- Still a selection-on-observables approach

Dale and Krueger Data

- DK's data come from College and Beyond (C&B) survey
  - Combination of college administrative data and survey data for students enrolled in 34 colleges in 1951, 1976, 1989
  - Sample of colleges is much more selective than the US average
  - Includes students' SAT scores, applications, college transcripts
  - DK (2002) focus on self-reported earnings in 1995 followup survey of students in the 1976 college cohort (mid/late 30s)
- DK (2014) match C&B to administrative earnings data from Social Security Administration (SSA) tax records through 2007
  - Better earnings information
  - Longer time horizon for 1976 cohort
  - Can also look at earnings for 1989 cohort

TABLE I
ILLUSTRATION OF HOW MATCHED-APPLICANT GROUPS WERE CONSTRUCTED

		Student applications to college							
		Application 1		Application 2		Application 3		Application 4	
Student	Matched-applicant group	School average SAT	School admissions decision	School average SAT	School admissions decision	School average SAT	School admissions decision	School average SAT	School admissions decision
Student A	1	1280	Reject	1226	Accept*	1215	Accept	na	na
Student B	1	1280	Reject	1226	Accept	1215	Accept*	na	na
Student C	2	1360	Accept	1310	Reject	1270	Accept*	1155	Accept
Student D	2	1355	Accept	1316	Reject	1270	Accept*	1160	Accept
Student E	2	1370	Accept*	1316	Reject	1260	Accept	1150	Accept
Student F	Excluded	1180	Accept*	na	na	na	na	na	na
Student G	Excluded	1180	Accept*	na	na	na	na	na	na
Student H	3	1360	Accept	1308	Accept*	1260	Accept	1160	Accept
Student I	3	1370	Accept*	1311	Accept	1255	Accept	1155	Accept
Student J	3	1350	Accept	1316	Accept*	1265	Accept	1155	Accept
Student K	4	1245	Reject	1217	Reject	1180	Accept*	na	na
Student L	4	1235	Reject	1209	Reject	1180	Accept*	na	na
Student M	5	1140	Accept	1055	Accept*	na	na	na	na
Student N	5	1145	Accept*	1060	Accept	na	na	na	na
Student O	No match	1370	Reject	1038	Accept*	na	na	na	na

* Denotes school attended.
na = did not report submitting application.
The data shown on this table represent hypothetical students. Students F and G would be excluded from the matched-applicant subsample because they applied to only one school (the school they attended). Student O would be excluded because no other student applied to an equivalent set of institutions.

TABLE V

LINEAR REGRESSIONS PREDICTING WHETHER STUDENT ATTENDED MOST SELECTIVE COLLEGE FOR C&B SAMPLE OF STUDENTS ADMITTED TO MORE THAN ONE SCHOOL

	Parameter estimates	
	Matched-applicant model*	Self-revelation model
Predicted log (parental income)	−0.024	−0.037
	(0.026)	(0.030)
Own SAT score/100	0.020	0.021
	(0.005)	(0.007)
Female	0.034	0.033
	(0.014)	(0.028)
Black	0.056	−0.005
	(0.026)	(0.037)
Hispanic	−0.019	0.042
	(0.064)	(0.074)
Asian	0.019	0.074
	(0.026)	(0.050)
Other/missing race	−0.095	0.010
	(0.093)	(0.081)
High school top 10 percent	−0.014	−0.020
	(0.021)	(0.028)
High school rank missing	−0.035	−0.040
	(0.036)	(0.058)
Athlete	0.056	0.059
	(0.023)	(0.045)
Average SAT score/100 of schools applied to		−0.122
		(0.040)
One additional application		0.149
		(0.037)
Two additional applications		0.076
		(0.033)
Three additional applications		0.020
		(0.038)
N	5536	8257

Only students who were accepted by more than one school are included in the sample.

Each equation also includes a constant term. Standard errors are in parentheses, and are robust to correlated errors among students who attended the same institution.

Equations are estimated by WLS; weights are designed to make the sample representative of the population of students at the C&B institutions.

* Applicants are matched by the average SAT score (within 25 point intervals) of each school at which they were accepted and rejected. Model includes 1,079 dummy variables indicating each set of matched applicants.

TABLE III
LOG EARNINGS REGRESSIONS USING COLLEGE AND BEYOND SURVEY, SAMPLE OF MALE AND FEMALE FULL-TIME WORKERS

	Model					
	Basic model: no selection controls		Matched-applicant model	Alternative matched-applicant models		Self-revelation model
	Full sample	Restricted sample	Similar school-SAT matches*	Exact school-SAT matches**	*Barron's* matches***	
Variable	1	2	3	4	5	6
School-average SAT score/100	0.076	0.082	−0.016	−0.106	0.004	−0.001
	(0.016)	(0.014)	(0.022)	(0.036)	(0.016)	(0.018)
Predicted log(parental income)	0.187	0.190	0.163	0.232	0.154	0.161
	(0.024)	(0.033)	(0.033)	(0.079)	(0.028)	(0.025)
Own SAT score/100	0.018	0.006	−0.011	0.003	−0.005	0.009
	(0.006)	(0.007)	(0.007)	(0.014)	(0.005)	(0.006)
Female	−0.403	−0.410	−0.395	−0.476	−0.400	−0.396
	(0.015)	(0.018)	(0.024)	(0.049)	(0.017)	(0.014)
Black	−0.023	−0.026	−0.057	−0.028	−0.057	−0.034
	(0.035)	(0.053)	(0.053)	(0.049)	(0.039)	(0.035)
Hispanic	0.015	0.070	0.020	−0.248	0.036	0.007
	(0.052)	(0.076)	(0.099)	(0.206)	(0.066)	(0.053)
Asian	0.173	0.245	0.241	0.368	0.163	0.155
	(0.036)	(0.054)	(0.064)	(0.141)	(0.049)	(0.037)
Other/missing race	−0.188	−0.048	0.060	−0.072	−0.050	−0.192
	(0.119)	(0.143)	(0.180)	(0.083)	(0.134)	(0.116)
High school top 10 percent	0.061	0.091	0.079	0.091	0.079	0.063
	(0.018)	(0.022)	(0.026)	(0.032)	(0.024)	(0.019)
High school rank missing	0.001	0.040	0.016	0.029	0.025	−0.009
	(0.024)	(0.026)	(0.038)	(0.066)	(0.027)	(0.022)
Athlete	0.102	0.088	0.104	0.169	0.093	0.094
	(0.025)	(0.030)	(0.039)	(0.096)	(0.033)	(0.024)
Average SAT score/100 of schools applied to						0.090
						(0.013)
One additional application						0.064
						(0.011)
Two additional applications						0.074
						(0.022)
Three additional applications						0.112
						(0.028)
Four additional applications						0.085
						(0.027)
Adjusted R^2	0.107	0.110	0.112	0.142	0.106	0.113
N	14,238	6,335	6,335	2,330	9,202	14,238

Each equation also includes a constant term. Standard errors are in parentheses and are robust to correlated errors among students who attended the same institution.

Equations are estimated by WLS and are weighted to make the sample representative of the population of students at the C&B institutions.

* Applicants are matched by the average SAT score (within 25 point intervals) of each school at which they were accepted or rejected. This model includes 1,232 dummy variables representing each set of matched applicants.

** Applicants are matched by the average SAT score of each school at which they were accepted or rejected. This model includes 654 dummy variables representing each set of matched applicants.

*** Applicants are matched by the *Barron's* category of each school at which they were accepted or rejected. This model includes 350 dummy variables representing each set of matched applicants.

Table 3

Comparing Parameter Estimates of the Effect of College Average SAT Score on Earnings Using C&B and SSA Data, 1976 Cohort

	C&B sample[a]		Merged C&B and SSA sample[b]									
	Log 1995 C&B earnings		Log 1995 C&B earnings		Log 1995 SSA earnings (topcoded)		Log 1995 SSA earnings (not topcoded)		Log (median of 1993 to 1997 earnings), SSA data		Log (median of 1993 to 1997 earnings), SSA data	
	1	2	3	4	5	6	7	8	9	10	11	12
	Basic	Self – revelation	Basic	Self - revelation	Basic	Self – revelation	Basic	Self - revelation	Basic	Self – revelation	Basic	Self – revelation
Parameter estimate for school SAT/100	0.076	-0.001	0.068	-0.007	0.048	-0.021	0.058	-0.015	0.059	-0.025	0.061	-0.023
	(.008)	(.012)	(.007)	(.012)	(.009)	(.014)	(.009)	(.015)	(.008)	(.012)	(.007)	(.012)
	{.016}	{.018}	{.014}	{.018}	{.016}	{.018}	{.017}	{.016}	{.012}	{.013}	{.013}	{.014}
N	14,238		10,886		10,886		10,886		11,932		12,075	
Sample restriction	Full-time workers (according to C&B survey)		Full-time workers (according to C&B survey)		Full-time workers (according to C&B survey)		Full-time workers (according to C&B survey)		Median earnings greater than zero (SSA data)		Median earnings greater than $13,822 in 2007 dollars (SSA data)	

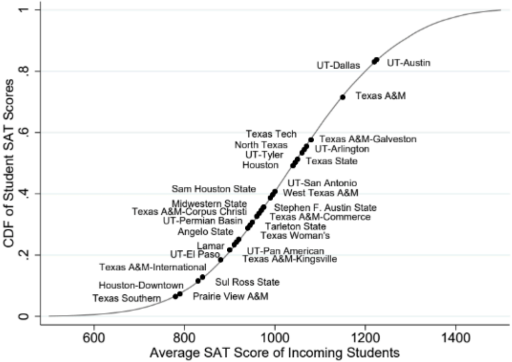
Table 5

Effect of School SAT Score/100 on Earnings, 1976 Cohort

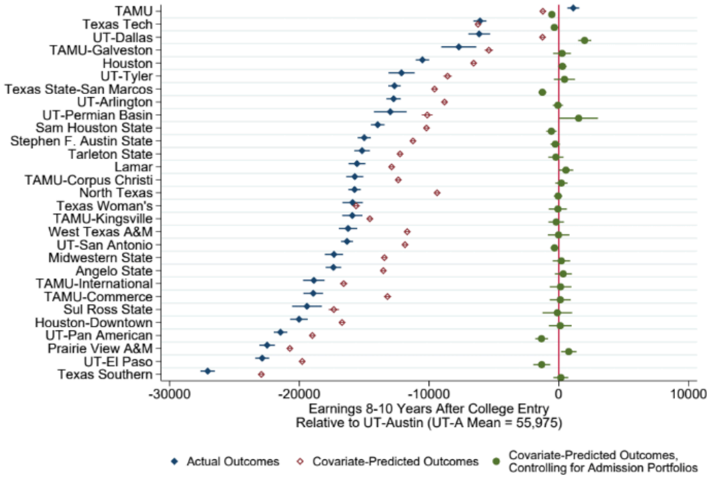
	Men		Women		Men and women pooled	
	Basic	Self-revelation	Basic	Self-revelation	Basic	Self-revelation
Effect on log (median of 1983 through 1987 earnings)						
Parameter estimate for school SAT/100	0.011	-0.004	-0.007	-0.037	0.004	-0.017
	(.007)	(.011)	(.007)	(.012)	(.005)	(.008)
	{.012}	{.019}	{.012}	{.023}	{.011}	{.014}
N	6,294	6,294	5,690	5,690	11,984	11,984
Effect on log (median of 1988 earnings through 1992 earnings)						
Parameter estimate for school SAT/100	0.054	-0.001	0.031	-0.034	0.045	-0.014
	(.009)	(.013)	(.009)	(.015)	(.006)	(.010)
	{.014}	{.016}	{.014}	{.019}	{.012}	{.013}
N	6,911	6,911	6,294	6,294	12,407	12,407
Effect on log (median of 1993 earnings through 1997 earnings)						
Parameter estimate for school SAT/100	0.080	0.001	0.034	-0.059	0.061	-0.023
	(.010)	(.016)	(.010)	(.018)	(.007)	(.012)
	{.013}	{.015}	{.012}	{.016}	{.013}	{.014}
N	6,896	6,896	5,179	5,179	12,075	12,075
Effect on log (median of 1998 earnings through 2002 earnings)						
Parameter estimate for school SAT/100	0.087	0.002	0.042	-0.069	0.070	-0.024
	(.012)	(.018)	(.012)	(.020)	(.008)	(.013)
	{.015}	{.022}	{.010}	{.016}	{.012}	{.019}
N	6,869	6,869	5,195	5,195	12,064	12,064
Effect on log (median of 2003 earnings through 2007 earnings)						
Parameter estimate for school SAT/100	0.083	0.006	0.057	-0.035	0.074	-0.008
	(.013)	(.020)	(.012)	(.021)	(.009)	(.014)
	{.015}	{.027}	{.013}	{.018}	{.014}	{.018}
N	6,650		5,244		11,894	

DK has been also replicated in Texas (w/ much better data) by Mountjoy and Hickman (2021)

Figure 1: Selectivity Across Colleges and SAT Score Variation Within Colleges



Mountjoy and Hickman (2021)



Notes: Each set of point estimates and robust 95% confidence intervals come from regressions of individual student outcomes on college treatment indicators (and cohort fixed effects), omitting UT-Austin as the reference treatment (signified by the vertical line at zero). The UT-Austin outcome mean appears in parentheses below each plot. *Covariate-Predicted Outcomes* replace actual outcomes with the predicted values from a separate OLS regression of each outcome on the following set of pre-college covariates: quartics in 10th grade math and reading scores, attendance, days of disciplinary suspension, and number of advanced high school courses; binary indicators for free/reduced price lunch eligibility, gender, at-risk status, and top decile GPA; and fixed effects for each race, high school cohort, and high school campus. The rightmost specification regresses these covariate-predicted outcomes on the college treatment indicators and solely controls for each distinct portfolio of college admissions (and cohort fixed effects).

Dale and Krueger Summary

- DK approach is built on comparisons of students who choose more- and less-selective colleges from the same set, for unknown reasons
- A natural story is that students with higher unobserved ability choose to matriculate to more selective schools. This would lead to *upward* bias in the comparison of more- and less-selective colleges
- Alternatively, perhaps money-conscious students choose the cheaper college option and also seek out higher-paying careers, leading to downward bias
- Nonetheless, DK's analysis shows that controlling for basic attributes of students' application portfolios is enough to kill the selective college advantage, a striking fact

schools may have higher unobserved ability. These results are consistent with the conclusion of Hunt's [1963, p. 56] seminal research: "The C student from Princeton earns more than the A student from Podunk not mainly because he has the prestige of a Princeton degree, but merely because he is abler. The golden touch is possessed not by the Ivy League College, but by its students."

The Careers of Young Men: Keane and Wolpin (JPE 1997)

- Now we'll shift gears and consider a structural model more tightly linked to the theory of human capital investment
- Keane and Wolpin (JPE 1997) estimate a dynamic model of schooling and occupational choice
- Forward-looking individuals choose how long to go to school and where to work
- Schooling and experience have heterogeneous effects on earnings by occupation
- Relies on strong parametric assumptions, but useful to understand the approach

Keane and Wolpin (1997): Notation

- At each age a between 16 and a terminal age A, an individual chooses between five alternatives:
  1. Blue collar job
  2. White collar job
  3. Military
  4. School
  5. Home production
- These alternatives are indexed by m
- The indicator $d_m(a)$ is one if the individual chooses alternative m at age a

Keane and Wolpin (1997): Notation

- $e_m(a)$: skill units (human capital) in alternative m at age a
- $x_m(a)$: experience in alternative m up to age a, $x_m(a) = \sum_{\ell=16}^{a} d_m(\ell)$
- $w_m(a)$: wage in alternative m at age a, $w_m(a) = r_m e_m(a)$
- r_m: return to human capital in alternative m
- $g(a)$: schooling by age a, $g(a) = \sum_{\ell=16}^{a} d_4(\ell)$

Keane and Wolpin (1997): Notation

- Specification for human capital:

$$e_m(a) = \exp\left(e_m(16) + e_{m1} g(a) + e_{m2} x_m(a) - e_{m3} x_m(a)^2 + \epsilon_m(a)\right)$$

- $e_m(16)$ is the initial endowment
- $\epsilon_m(a)$ is unobserved time-varying heterogeneity
- KW assume $\epsilon_m(a) \sim$ iid $N(0, \Omega)$

Keane and Wolpin (1997): Notation

- Flow utility ("reward") from working in occupation m at age a:

$$R_m(a) = r_m \cdot \exp\left(e_m(16) + e_{m1}g(a) + e_{m2}x_m(a) - e_{m3}x_m(a)^2 + \epsilon_m(a)\right)$$

- Utilities for schooling and home production:

$$R_4(a) = e_4(16) - tc_1 \cdot 1\{g(a) \geq 12\} - tc_2 \cdot 1\{g(a) \geq 16\} + \epsilon_4(a)$$

$$R_5(a) = e_5(16) + \epsilon_5(a)$$

Value Function

- The vector of state variables is

$$S(a) = (e(16), x(a), g(a), \epsilon(a))$$

- $S(a)$ only includes the current-period shock; no history of $\epsilon(a)$
- Value function at age a:

$$V(S(a), a) = \max_m V_m(S(a), a)$$

$$V_m(S(a), a) = R_m(S(a), a) + \delta E\left[V(S(a+1), a+1) | S(a), d_m(a) = 1\right] \text{ for } a < A$$
$$V_m(S(A), A) = R_m(S(A), A)$$

- Evolution of state variables:

$$x_m(a+1) = x_m(a) + d_m(a)$$

$$g(a+1) = g(a) + d_4(a)$$

Solution

- In principle the optimization problem can be solved by backward recursion:
  1. Begin in period A. For all possible $S(A)$, calculate the best choice $d^*(S(A), A)$ and corresponding $V(S(A), A)$
  2. Move to period $A-1$. For every m and every $S(A-1)$, use the results from (1) to compute $E[V(S(A), A)|S(A-1), d_m(A-1) = 1]$. From these values, construct $V_m(S(A-1), A-1)$.
  3. Using the results from (2), maximize over m to calculate $d^*(S(A-1), A-1)$ and $V(S(A-1), A)$ for every value of $S(A-1)$.
  4. Continue to move backwards, repeating steps (2) and (3) to compute the value function in every period
- In practice the state space is too large – KW use an interpolation method after computing the value function at a subset of points in the state space

Taming the Curse of Dimensionality: Quantitative Economics with Deep Learning
Jesús Fernández-Villaverde, Galo Nuño, and Jesse Perla
NBER Working Paper No. 33117
November 2024
JEL No. C61, C63, E27

ABSTRACT

We argue that deep learning provides a promising avenue for taming the curse of dimensionality in quantitative economics. We begin by exploring the unique challenges posed by solving dynamic equilibrium models, especially the feedback loop between individual agents' decisions and the aggregate consistency conditions required by equilibrium. Following this, we introduce deep neural networks and demonstrate their application by solving the stochastic neoclassical growth model. Next, we compare deep neural networks with traditional solution methods in quantitative economics. We conclude with a survey of neural network applications in quantitative economics and offer reasons for cautious optimism.

Jesús Fernández-Villaverde
Department of Economics
University of Pennsylvania
The Ronald O. Perelman Center
for Political Science and Economics
133 South 36th Street Suite 150
Philadelphia, PA 19104
and CEPR
and also NBER
jesusfv@econ.upenn.edu

Galo Nuño
Banco de España
Bank of Spain
Spain
galonunobarrau@gmail.com

Jesse Perla
Vancouver School of Economics
University of British Columbia
jesse.perla@ubc.ca

Estimation

- Parameters to be estimated: Returns to schooling and experience, schooling costs, covariance matrix of $\epsilon(a)$, initial endowment $e(16)$
- Observed outcomes at age a are $c(a) = \{d_m(a), w_m(a)\}$ for alternatives 1-3 and $c(a) = \{d_m(a)\}$ for alternatives 4-5
- Define $\bar{S}(a) = \{e(16), g(a), x(a)\}$, the state vector excluding unobserved shocks
- Probability of an individual's choice sequence is

$$Pr\left[c(16), ..., c(A)|g(16), e(16)\right] = \prod_{a=16}^{A} Pr[c(a)|\bar{S}(a)]$$

- Solution to the optimization problem is used to compute the choice probabilities
- Estimate by maximum simulated likelihood – no closed form

Finite Types

- Initial skill endowment $e(16)$ is likely to be heterogeneous
- KW allow for a set of K unobserved types. New likelihood function:

$$Pr\left[c(16),...,c(A)|g(16),e(16)\right] = \sum_{k=1}^{K}\prod_{a=16}^{A}\pi_{k|g(16)}Pr[c(a)|\bar{S}(a),\text{type}=k]$$

- Type proportions are added to vector of parameters to be estimated
- "Finite types" approach is a common approach to modeling unobserved heterogeneity in a tractable way

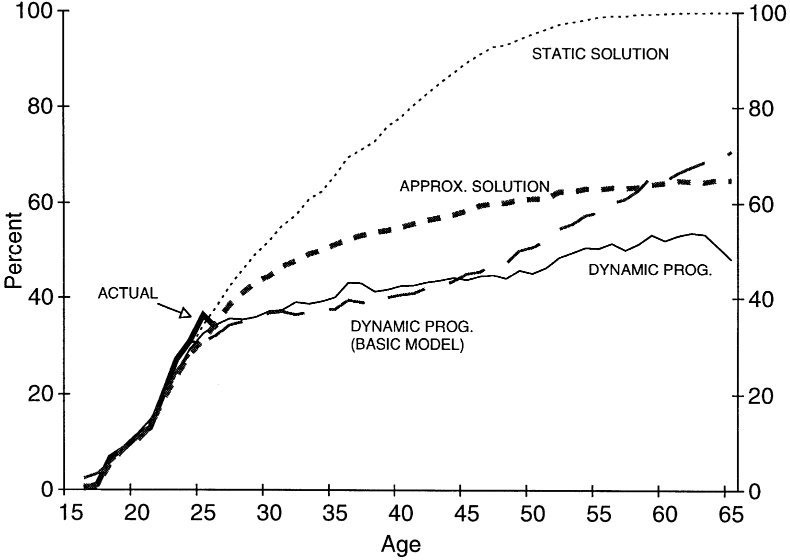


FIG. 1.—Percentage white-collar by age

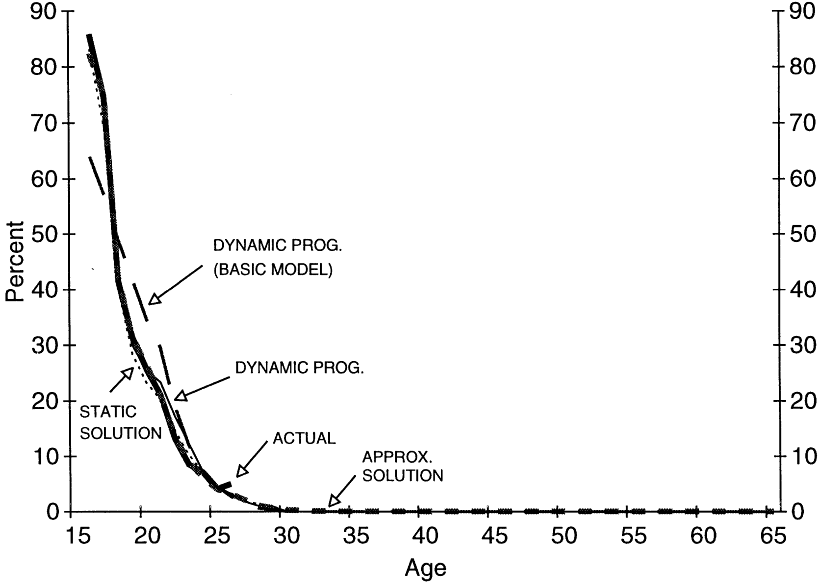


FIG. 4.—Percentage in school by age

Extended Model

- Basic model fails to fit some important features of the data (e.g. school attendance, persistence in occupational choice, Mincer coefficients)
- KW estimate an "extended" model adding:
    - Extra terms in human capital production function (skill depreciation, age effects, high school/college graduation effects)
    - Costs for switching occupations/re-entering school
    - Non-pecuniary tastes for occupations
- Use model to consider counterfactuals (college tuition subsidy).

TABLE 15

DISTRIBUTIONAL EFFECTS OF A $2,000 COLLEGE TUITION SUBSIDY

	Type 1	Type 2	Type 3	Type 4
Mean expected present value of lifetime utility at age 16:				
No subsidy	413,911	391,162	225,026	286,311
Subsidy	419,628	392,372	226,313	288,109
Gross gain	5,717	1,210	1,287	1,798
Net gain:				
Subsidy to all types*	3,513	−994	−917	−406
Subsidy to types 2, 3, and 4†	−1,134	76	153	664
Subsidy to types 3 and 4‡	−862	−862	425	936

* The per capita cost of the subsidy program is $2,204.
† The per capita cost of the subsidy program is $1,134.
‡ The per capita cost of the subsidy program is $862.

Signaling

- Evidence so far suggests schooling increases earnings
- Conventional human capital view that we have viewed thus far is that schooling investments raise wages by boosting productivity
- Signaling models (Spence, 1973) provide an alternative explanation.
- Idea: is that schooling provides a signal to firms that an individual is a "high-ability" type.

Signaling Evidence

- Evidence on labor market signaling is hard to come by
- Ideal data for identifying the returns to a signal would contain exogenous variation in signaling status among individuals with similar levels of human capital.
    - In other words, need workers with identical productivities but different values of an observed credential – an instrument for schooling won't do it

Clark and Martorell (2014)

- Clark and Martorell (JPE 2014) estimate the signaling value of a high school diploma.
- OLS return to education is especially large for grade 12
- How much of this "sheepskin effect" reflects the signaling value of the diploma?

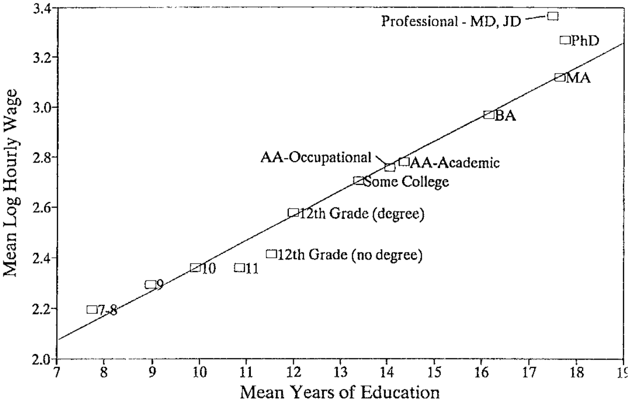


Fig. 2. Relationship between mean log hourly wages and completed education, men aged 40–45 in 1994–1996 Current Population Survey. Mean education by degree category estimated from February 1990 CPS.

TABLE 6
ASSOCIATIONS BETWEEN DIPLOMA AND TEST SCORES AND EARNINGS

A. MEAN DIFFERENCES BY DIPLOMA STATUS

	Last-Chance Sample (1)	Complete Grade 12, No College: All (2)	T1 (3)	T2 (4)	T3 (5)
Earnings years 7–11	1,814.7	2,867.8	1,780.3	1,752.0	2,385.3
	(138.1)	(79.3)	(111.8)	(176.1)	(228.5)
Observations	128,460	992,031	210,793	193,970	194,896
Mean earnings without diploma	12,400	12,673	11,858	13,301	13,538
Difference (%)	14.6	22.6	15.0	13.2	17.6
PDV earnings	8,054.5	8,731.0	7,280.7	7,459.4	10,546.3
	(632.3)	(341.9)	(501.9)	(779.8)	(951.4)
Observations	37,571	340,028	74,490	63,652	64,548
Mean earnings without diploma	70,280	69,992	66,466	74,216	73,860
Difference (%)	11.5	12.5	11.0	10.1	14.3

B. CORRELATION BETWEEN TEST SCORE AND PDV EARNINGS IN LAST-CHANCE SAMPLE

	(1)	(2)	(3)	(4)
Last-chance score	870.6	983.2	968.8	962.7
	(40.0)	(60.6)	(61.0)	(103.6)
Last-chance score^2		11.3	14.3	14.1
		(3.8)	(7.9)	(7.4)
Last-chance score^3			.2	.2
			(.3)	(1.0)
Last-chance score^4				.0
				(.0)
Polynomial degree	1	2	3	4
p-value slopes = 0	.00	.00	.00	.00
R^2	.0126	.0129	.0129	.0129
Adjusted R^2	.0126	.0128	.0128	.0128

NOTE.—Panel A shows mean differences in earnings by high school diploma status. T1, T2, and T3 refer to bottom, second-bottom, and third-bottom tertiles of the ability distribution as measured by initial exam scores. No restrictions are placed on the last-chance sample (i.e., we do not restrict to those with no college). Panel B shows estimates of a regression of PDV earnings through year 11 ($r = .05$) on a polynomial in the last-chance exam score (each column represents a separate regression) for students in the last-chance sample. All models are estimated with no additional covariates beyond those listed in the table (high school diploma in panel A and the test score polynomial terms in panel B) and an intercept.

Clark and Martorell (2014)

- CM use the fact that students in Texas must pass exams before graduating high school
- Testing starts in 10th grade and students can try multiple times, but eventually face a "last chance" exam at the end of 12th grade
- CM combine data on last chance exams with UI earnings data up to 11 years after graduation, and estimate the effect of diploma receipt in an RD design
- There is some "slippage" even with last-chance exams – so the RD is fuzzy

Clark and Martorell (2014)

- 2SLS system:

$$Y_i = \beta_0 + \beta_1 D_i + g\left(p_i; \gamma_2\right) + \epsilon_i$$

$$D_i = \alpha_0 + \alpha_1 PASS_i + g(p_i; \gamma_1) + \omega_i$$

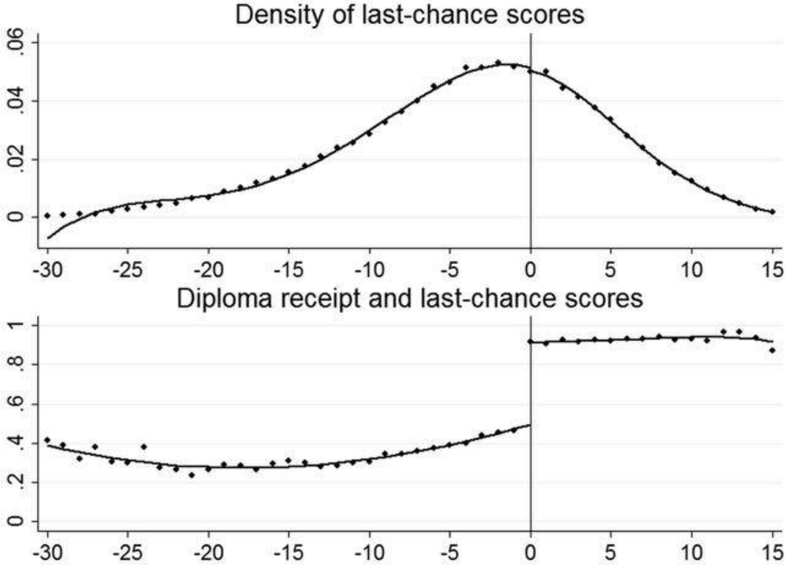


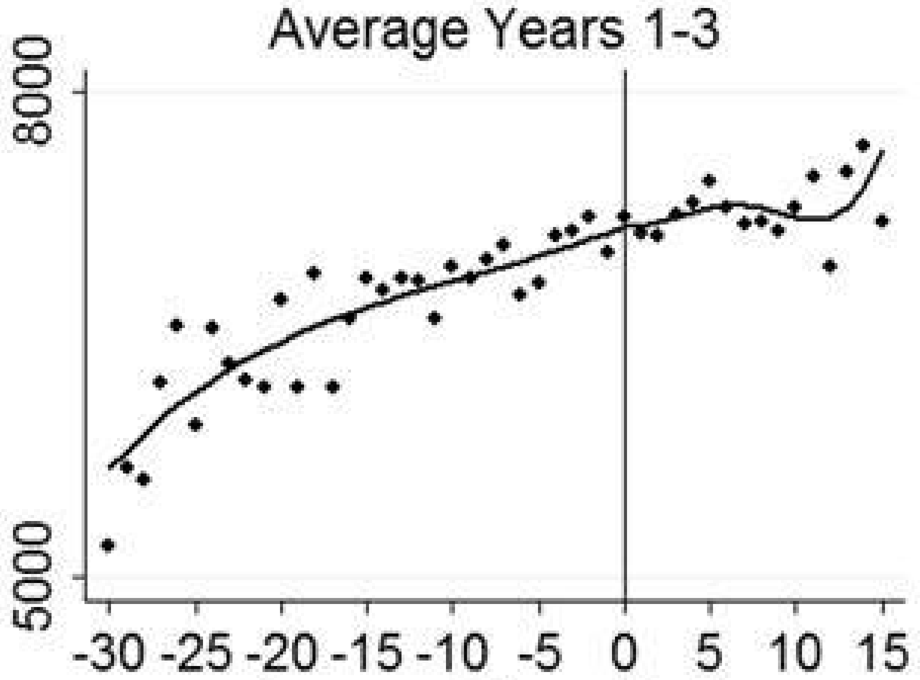
FIG. 1.—Last-chance exam scores and diploma receipt. The graphs are based on the last-chance sample. See table 1 and the text. Dots are test score cell means. The scores on the x-axis are the minimum of the section scores (recentered to be zero at the passing cutoff) that are taken in the last-chance exam. Lines are fourth-order polynomials fitted separately on either side of the passing threshold.

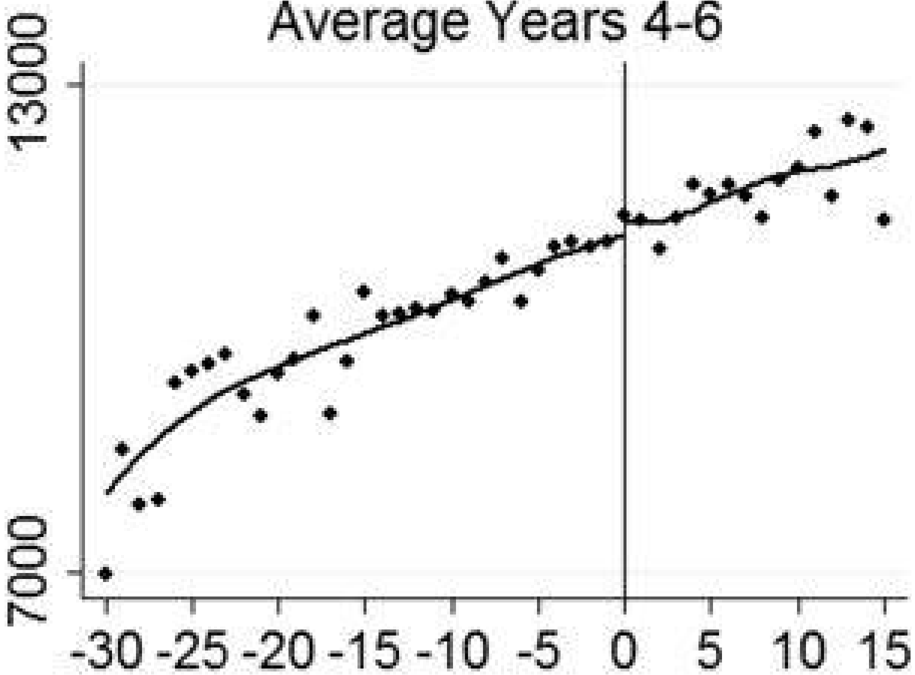
TABLE 2

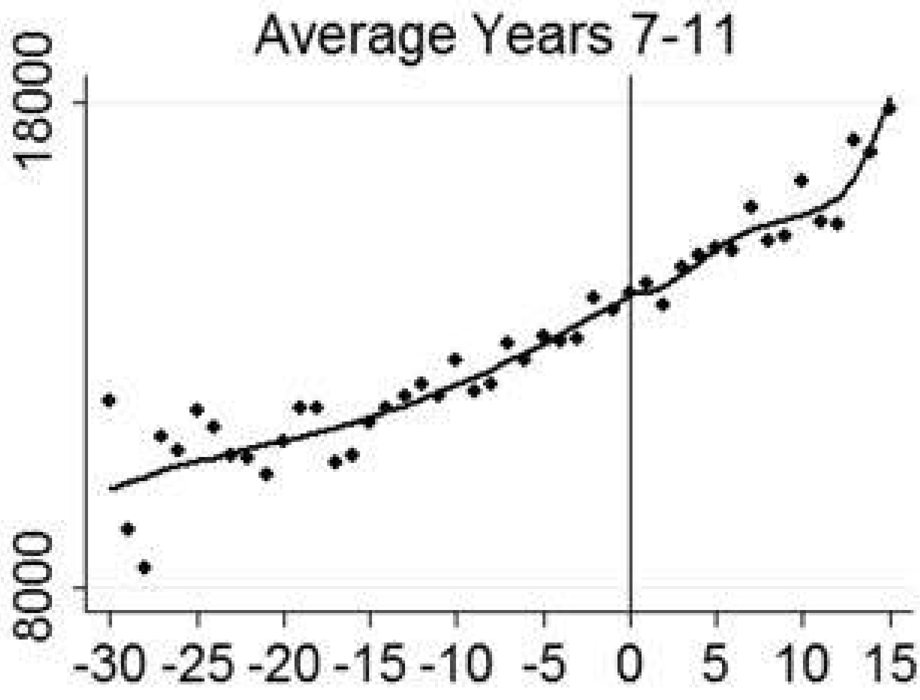
IMPACT OF PASSING THE LAST-CHANCE EXAM ON THE PROBABILITY OF EARNING A DIPLOMA

Receive High School Diploma	(1)	(2)	(3)	(4)	(5)
By end of summer after 12th grade (sample mean = .363)	.545	.484	.481	.475	.486
	(.007)	(.009)	(.012)	(.016)	(.009)
Within 1 year of last-chance exam (sample mean = .452)	.480	.420	.425	.424	.422
	(.007)	(.009)	(.012)	(.016)	(.009)
Within 2 years of last-chance exam (sample mean = .465)	.472	.415	.419	.417	.417
	(.007)	(.009)	(.012)	(.016)	(.009)
Within 3 years of last-chance exam (sample mean = .468)	.468	.412	.416	.414	.414
	(.007)	(.009)	(.012)	(.016)	(.009)
Baseline covariates?	No	No	No	No	Yes
Degree of test score polynomial	1	2	3	4	2

NOTE.—The table is based on last-chance samples (see table 1 and the text). "Degree of test score polynomial" refers to the test score polynomials controlled for in these regressions (all interacted with a dummy for passing the exam). Column 5 presents estimates based on models that also control for covariates (see note to table 1). Robust standard errors are in parentheses. There are 37,571 observations in each panel.


Average Years 1-3
8000
5000
-30 -25 -20 -15 -10 -5 0 5 10 15


Average Years 4-6
13000
7000
-30
-25
-20
-15
-10
-5
0
5
10
15


Average Years 7-11
18000
8000
-30
-25
-20
-15
-10
-5
0
5
10
15

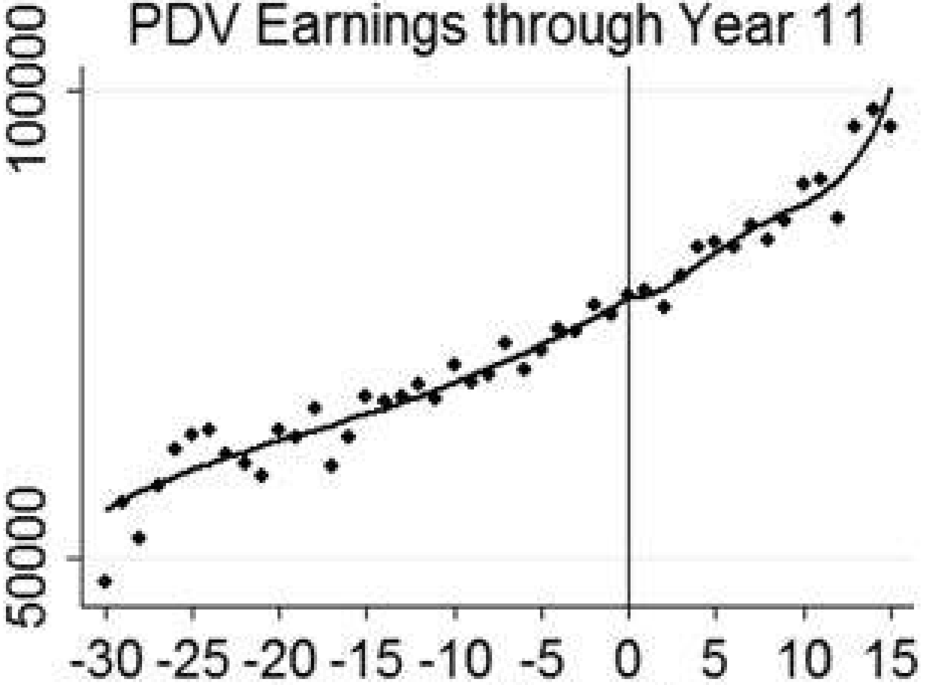

PDV Earnings through Year 11
100000
50000
-30
-25
-20
-15
-10
-5
0
5
10
15

TABLE 4

IMPACT OF RECEIVING A HIGH SCHOOL DIPLOMA ON EARNINGS

	(1)	(2)
Years 1–3	109.5	119.8
	(352.0)	(283.2)
	1.6%	1.7%
Years 4–6	333.1	155.0
	(517.9)	(421.7)
	3.0%	1.4%
Years 7–11	99.3	52.2
	(767.0)	(634.0)
	.7%	.4%
All years pooled	177.7	106.4
	(475.9)	(392.9)
	1.7%	1.0%
PDV earnings through year 11	1,632.1	598.8
	(3,299.5)	(2,742.3)
	2.1%	.8%
Polynomial specification	Unrestricted	Restricted

NOTE.—Estimates use the same samples and specification as cols. 5 and 6 of table 3. Unrestricted specification estimates are from two-stage least-squares models in which the test score polynomial is fully interacted with an indicator for passing the last-chance exam. Results from the "restricted" polynomial specification are the ratio of the reduced-form and first-stage estimates, where these are estimated separately and the standard errors are calculated using the delta method. The reduced form is the estimated discontinuity in earnings using the polynomial specification in which the slopes are constrained to be equal on either side of the passing cutoff (col. 6 in table 3) and the first stage is estimated using a polynomial that is fully interacted with the passing dummy. All standard errrors are adjusted for clustering at the individual level. For each set of estimates, the third row represents the point estimate expressed as a percentage of mean earnings just to the left of the passing cutoff.

TABLE 5
SIGNALING VALUES OF A HIGH SCHOOL DIPLOMA FOR SUBGROUPS

	By Gender		By Race		By County: Mobility and Federal Employment	
	Male	Female	White	Nonwhite	Low	High
Years 1–3	188.1	91.1	−8.9	154.1	137.9	89.7
	(546.8)	(303.6)	(561.1)	(322.6)	(393.7)	(407.4)
Years 4–6	452.0	8.7	−228.2	247.0	556.0	−276.6
	(814.2)	(449.9)	(838.7)	(480.9)	(584.9)	(607.5)
Years 7–11	492.0	−138.6	−124.4	19.8	−149.7	329.1
	(1,224.5)	(655.6)	(1,269.6)	(703.3)	(878.2)	(916.1)
Years pooled	383.5	−17.7	−120.7	135.8	163.7	57.3
	(765.5)	(407.2)	(791.3)	(438.5)	(551.6)	(557.7)
PDV earnings through year 11	2,473.8	−297.0	938.5	−1,457.0	1,037.9	114.7
	(5,391.4)	(2,824.2)	(3,035.7)	(5,621.4)	(3,885.9)	(3,849.0)

NOTE.—Estimates are from the preferred restricted polynomial specification with baseline covariates. See the text and note to table 4 for additional details.

Clark and Martorell (2014): Results

- CM show that diploma receipt has no effect on earnings. Does this disprove the signaling view of education?

Clark and Martorell (2014): Results

- CM show that diploma receipt has no effect on earnings. Does this disprove the signaling view of education?
  - Does the diploma capture information about productivity?
  - Do employers observe diploma status?
  - Can firms observe productivity information for this population from other sources?
- Consensus seems to be that empirical support for signaling theory is scant (Deming, 2022).

References

- Angrist, Joshua and Alan Krueger (1991). "Does Compulsory Schooling Attendance Affect Schooling and Earnings?" Quarterly Journal of Economics 106 (4).
- Arcidiacono, Peter, Patrick Bayer, and Aurel Hizmo (2010). "Beyond Signaling and Human Capital: Education and the Revelation of Ability." American Economic Journal: Applied Economics 2 (4).
- Ashenfelter, Orley, and Cecilia Rouse (1998). "Income, Schooling and Ability: Evidence from a New Sample of Twins." Quarterly Journal of Economics 113 (1).
- Bedard, Kelly (2001). "Human Capital Versus Signaling Models: University Access and High School Dropouts." Journal of Political Economy 109 (4).
- Ben-Porath, Yoram (1967). "The Production of Human Capital Over the Life Cycle." Journal of Political Economy 75 (4).
- Bound, John, David Jaeger, and Regina Baker (1995). "Problems with Instrumental Variables Estimation when the Correlation Between the Instruments and the Endogenous Explanatory Variable is Weak." Journal of the American Statistical Association 90 (430).
- Becker, Gary (1993). Human Capital. Chicago: University of Chicago Press.
- Buckles, Kasey, and Daniel Hungerman (2013). "Season of Birth and Later Outcomes: Old Questions, New Answers." Review of Economics and Statistics 95 (3).

References

- Cameron, Stephen, and Christopher Taber (2004). "Estimation of Educational Borrowing Constraints Using Returns to Schooling." Journal of Political Economy 112 (1).
- Card, David (1999). "The Causal Effect of Education on Earnings." Handbook of Labor Economics, Volume 3A.
- Card, David (2001). "Estimating the Return to Schooling: Progress on Some Persistent Econometric Problems." Econometrica 69 (5).
- Carneiro, Pedro, and James Heckman (2002). "The Evidence on Credit Constraints in Post-Secondary Schooling." The Economic Journal 112.
- Clark, Damon, and Paco Martorell (2014). "The Signaling Value of a High School Diploma." Journal of Political Economy 122 (2).
- Cunha, Flavio, and James Heckman (2007). "The Technology of Skill Formation." American Economic Review 97 (2).
- Dale, Stacy, and Alan Krueger (2002). "Estimating the Payoff to Attending a More Selective College: An Application of Selection on Observables and Unobservables." Quarterly Journal of Economics 117 (4).
- Dale, Stacy, and Alan Krueger (2014). "Estimating the Effects of College Characteristics Over the Career Using Administrative Earnings Data." Journal of Human Resources 49 (2).
- Devereux, Paul, and Robert Hart (2010). "Forced to be Rich? Returns to Compulsory Schooling in Britain." The Economic Journal 120 (54).

References

- Duflo, Esther (2001). "Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment." American Economic Review 91 (4).
- Grenet, Julien (2013). "Is Extending Compulsory Schooling Alone Enough to Raise Earnings? Evidence from French and British Compulsory Schooling Laws." Scandinavian Journal of Economics 115 (1).
- Griliches, Zvi (1977). "Estimating the Returns to Schooling: Some Econometric Problems." Econometrica 45 (1).
- Heckman, James, and Edward Vytlacil (1998). "Instrumental Variables Methods for the Correlated Random Coefficient Model: Estimating the Rate of Return to Schooling When the Return is Correlated with Schooling." Journal of Human Resources 33 (4).
- Keane, Michael, and Kenneth Wolpin (1997). "The Career Decisions of Young Men." Journal of Political Economy 105 (3).
- Lang, Kevin, and David Kropp (1986). "Human Capital Versus Sorting: The Effects of Compulsory Attendance Laws." Quarterly Journal of Economics 101 (3).
- MaCloud, W. Bentley, Evan Riehl, Juan E. Saavedra, and Miguel Urquiola (forthcoming). "The Big Sort: College Reputation and Labor Market Outcomes." American Economic Journal: Applied Economics.

References

- Oreopolous, Philip (2006). "Estimating Average and Local Average Treatment Effects of Education when Compulsory Schooling Laws Really Matter." American Economic Review 96 (1).
- Spence, Michael (1973). "Job Market Signaling." Quarterly Journal of Economics 87 (3).
- Stiglitz, Joseph (1975). "The Theory of 'Screening,' Education, and the Distribution of Income." American Economic Review 65 (3).
- Tyler, John, Richard Murnane, and John Willett (2000). "Estimating the Labor Market Signaling Value of the GED." Quarterly Journal of Economics 115 (2).
- Zimmerman, Seth (2014). "The Returns to College Admission for Academically Marginal Students." Journal of Labor Economics 32 (4).